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QUANTIFYING CIVILIAN OVERSIGHT: FREE-TEXT ANALYSIS OF CIVILIAN COMPLAINTS AGAINST ALBUQUERQUE POLICE

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M.SC. APPLIED SOCIAL DATA SCIENCE

Declaration

I hereby declare that this Thesis is entirely my own work and that it has not been submitted as an exercise for a degree at this or any other university.

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Abstract

Mitigating police misconduct in the United States via external regulation of law enforcement is a critical though formidable venture. Despite being touted by reformers, Civilian Review Boards (CRBs), in their varied forms, have produced unclear evidence of their effectiveness in either constraining police power or improving citizen-officer trust. Exploring the content that flows in and out of a single CRB would provide key insights to the process of civilian oversight, its strengths and weaknesses, and highlight potential places for improvement in the process. Advancing the use of Natural Language Processing (NLP) tools in the field of police science, this project employs sentence embeddings to assign binary signals across the Albuquerque Civilian Police Oversight Agency's (CPOA) corpus of finding letters from investigations into civilian complaints published between 2018 and 2025. Descriptive binary variables are tested for significance in predicting "sustained" findings against officers of the Albuquerque Police Department, with contextually informative variables performing the best. While the results are interpreted with caution given mitigating factors of the embeddings' moderate accuracy performance, the data presents a variety of shortcomings by the CPOA. Wielding a nuanced understanding of the American police force, this thesis takes a pioneering step toward bringing the NLP tools used for "smart policing" to the field of external law enforcement regulation.

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Nomenclature

X	Independent/input/predictor variable
Y	Outcome variable
β	Coefficients indicating contribution of predictor variables to the log odds of the outcome variable
χ^2	Coefficient indicating covariance
APD	Albuquerque Police Department
BCSO	Bernalillo County Sheriff's Office
(S)BERT	(Sentence)-Bidirectional Encoder Representations from Transformers
CD	Consent Decree (federal)
CPOA	Civilian Police Oversight Agency
CRB	Civilian Review Board
DoJ	United States Department of Justice
IAB	Internal Affairs Bureau
IRO	Internal Review Office
MH	Mental Health
NB	Naive Bayes (classifier)
NLP	Natural Language Processing
OBRD	On-Body Recording Device
OCR	Optical Character Recognition
SVM	Support Vector Machine (classifier)
TF-IDF	Term-Frequency Inverse Document Frequency (weighting, encoding)
UOF	Use of Force

1 Introduction

Civilian Review Boards (CRBs) have emerged as the popular choice in regulating American police misconduct and simultaneously improving police-community relationships (Office of Justice Programs, 2016). Though mechanisms for external constraint have been called for since the inception of the American police force (Prenzler, 2000), CRBs reached particular popularity following the Obama administration's publication of a task force report on 21st century policing, which highlighted it as a necessary mechanism for maintaining community-officer trust and ensuring misconduct is addressed (President's Task Force on 21st Century Policing, 2015). Public demands for CRBs, among other reform measures, continue to flare after each high-profile incident of officer misconduct or use of extreme force (Dlugolenski et al., 2024).

While some studies have investigated the effectiveness of CRBs in ameliorating strains on police-civilian relations (Schiff et al., 2025; Worden et al., 2018), the ability of a CRB to conduct impartial review of police incidents and curb misconduct remains unverified in the literature (Mugari, 2023). As a human institution, it is natural for biases to leak into even the most "impartial" organizations. The mere existence of a CRB that proclaims its impartiality is not enough to prove that police misconduct is in check. Investigating the actual content flowing in and out of these organizations is the only practical way to examine their activity for evidence of partial treatment of either police or civilian complainants. Most of the content published by CRBs, however, is either aggregated seasonal reports or in unstructured textual format (City of Albuquerque, n.d.).

Rising to the challenge of operationalizing unstructured texts as data, NLP tools are being increasingly tested in the field of police science. Many studies seeking to quantify law enforcement records for the aim of "smart policing" have applied NLP methodologies with some very promising results (Lukmanjaya et al., 2026). Naive Bayes (NB), Latent Dirichlet Allocation (LDA), Bidirectional Encoder Representations from Transformers (BERT), and other algorithms have been used to track instances of domestic violence (Hodgkinson et al., 2023), extract the mention of mental health conditions in police narratives (Karystianis et al., 2018), and more. These tools have largely yet to be employed in answering the demand for investigating CRBs (President's Task Force on 21st Century Policing, 2015;

Mugari, 2023), however, and doing so would be an essential step in the development of NLP in police science.

Part of the challenge of investigating and comparing CRB materials across contexts is the wide variation in their structure and powers (Filstad and Gottschalk, 2010). Honing in on one particular CRB allows an in-depth analysis of the textual data. The City of Albuquerque is a prime case for this purpose. A lawsuit brought against the city by the Department of Justice (DoJ) in 2014 cited Albuquerque Police Department's (APD) excessive use of force (UOF) as a violation of citizens' rights. The settlement reached between the DoJ and the city required the establishment of a CRB called the "Civilian Police Oversight Agency" (CPOA) (City of Albuquerque, 2023), a slew of amendments to APD policy, and a period of federal oversight (2014−2025) lasting until benchmarks of improvement were met (U.S. Department of Justice, 2014). The presence of federal oversight in this case promoted transparent data reporting on behalf of the CPOA, and created multiple sources of metrics on APD misconduct, making it ideal for my exploratory study.

In the following report, I will show how I developed a dataset on civilian complaints against APD processed by the CPOA to qualify the contents of their activities as a CRB. The beginning will describe the social context of this issue, underlining the importance of my specific inquiry. Next, I will move to explaining the CPOA's organizational structure and the data generation process of citizen complaint finding letters. The consequent methodology section will report how I tested the use of a variety of NLP tools in predicting the presence of my parameters of interest with moderate success. I will then discuss the results of my descriptive statistical analyses and multivariable logistic regression models predicting the output variable of "sustained" findings against APD employees. The penultimate chapter explores my analysis of the data, highlighting the preliminary evidence of the CPOA harboring pro-police sentiments in their handling of civilian complaints. The report concludes with recommended measures to be taken for mitigating such biases and a discussion of future directions in the research on civilian oversight of police in Albuquerque and beyond.

2 | Police Regulation in the U.S.

The endeavor to establish reliable external regulation of police organizations in the United States is uniquely important given the potent risk of corrupt and biased practices that violate citizens' rights. The record of American police forces as a racially and socio-economically biased institution is well-documented in the literature (Alexander, 2010; De Orio, 2022; Parks and Kirby, 2021). From federal policies resulting in discriminatory policing practices, to the monetization of the prisoner via the private prison industrial complex, institutional racial biases at all levels have precipitated severe disparities in police use of force against Black Americans (Alexander, 2010). The discrimination is not only racial, but social and economic—minority identity groups like the LGBTQIA+ (especially transgender citizens) and Americans living under the poverty line have long experienced police brutality at higher rates than the majority (Panfil, 2026; De Orio, 2022). In addition to committing violent enactments of racial and socioeconomic prejudices, corrupt American police have engaged in quasi-legal or illegal activities for their personal or financial benefit, a trend chronicled in research since at least the 1990s (Prenzler, 2000). Police corruption might manifest simply as preferential treatment of family or friends or, more disconcertingly, as direct involvement with criminal enterprises (Arnold, 2025). A police force committing these crimes historically cannot be trusted to self-regulate. Further, mechanisms of the American justice system below the level of costly federal intervention tend to fail in regulating police power due to their organizational proximity to law enforcement (Arnold, 2025). Even in the rare case that police misconduct reaches the desk of a district attorney, they may be reluctant to bring charges against members of the organization they collaborate with on a near-daily basis (Arnold, 2025). This has precipitated myriad efforts to control police agencies externally, including the promotion of the Civilian Review Board (CRB) (Prenzler, 2000).

2.1 Challenges in Evaluating Civilian Review Boards

CRBs have risen to popularity for their potential in establishing external constraint on police (President's Task Force on 21st Century Policing, 2015; Keyes and Keyes, 2023). Often formed as a

stipulation of a federal consent decree (CD), CRBs are groups of citizens living in the policed area appointed by local government to conduct independent investigations of complaints raised against police, UOF incidents, and other cases of potential misconduct (Schiff et al., 2025; Arnold, 2025). CRBs' power over law enforcement agencies differ, but structured independence and the ability to recommend disciplinary action are endorsed qualities of effective CRBs (Filstad and Gottschalk, 2010). The main aims of the CRB are practical and ideological: providing direct outsider control *and* improving community trust in police by bolstering the belief that misconduct will be punished (Ferdik et al., 2013). As strong a premise on which they stand, the challenge of assessing the effectiveness with which CRBs achieve either of those goals plagues our efforts to provide clear data-driven support for the continued investment in this form of police oversight.

Barriers in quantifying and comparing the material outputs of CRBS and CDs alike has prohibited clear consensus on their long-term positive impacts (Alpert et al., 2017; Mugari, 2023). A great portion of the literature is dedicated to operationalizing CRB success in increasing community trust in the police (Schiff et al., 2025; Worden et al., 2018), though often yielding inconsistent findings (Keyes and Keyes, 2023). Even efforts to prove the sustained reform of officer conduct after the resolution of a CD have found mixed results (Alpert et al., 2017). Beyond the measurement of CRB compliance with CDs during periods of federal oversight, little effort has been expended on evaluating CRB's impartiality or their practical impact on police disciplinary outcomes. This could be because we lack clarity on the nature of cases being seen by CRBs and consistency in our expectations of their investigatory processes. Given the variety of CRB structures and scopes of authority (and likewise local police jurisdictions and powers), comparison of CRBs across contexts faces many additional hurdles. Delving into one specific CRB, however, and quantifying its input and output with consideration for its unique social and organizational context would color our understanding of the CRB. This could even support the development of a framework for assessing and improving CRBs. To such an end, the following project utilizes tools from the field of Natural Language Processing (NLP) to manipulate previously untapped sources of free-text data provided by Albuquerque's CRB, the CPOA.

2.2 Natural Language Processing and Police Science

The field of data science has been leveraged in an effort to advance what is called "smart policing" (Sarzaeim et al., 2023). "Smart policing" seeks to utilize past records of criminal activity to predict a variety of phenomena linked to criminality, including the social web of criminal networks (Crandell and Korkmaz, 2018) and criminal profiles (Sarzaeim et al., 2023). The application of Natural Language Processing (NLP) tools in such contexts as predicting risk for recurrent domestic violence incidents (Karystianis et al., 2018), crime response risk levels (Baek et al., 2021), or risk of traffic collisions

(Bäumler and Prokop, 2024) exemplify the power of operationalizing free-text data in law enforcement.

The promising performance of NLP tools in police science thus far has paved the way for an inquiry analyzing civilian oversight unstructured textual data. Lukmanjaya et al. (2026) conducted a global review of text analysis studies that draw from texts relating to law enforcement encounters (police reports, case narratives, first-response call logs, etc.), with many machine learning classifiers reaching F1 scores above 0.7. For example, Hodgkinson et al. (2023) utilized the pre-trained Bidirectional Encoder Representations from Transformers (BERT) algorithm to isolate distinct domestic violence events from police reports such that aggregation occurring in the report did not skew total estimation of incidents. Their model, trained on 2.7% of the data (1200 out of 43488 samples), reaped an F1 score of 0.84 on the held-out test set (Hodgkinson et al., 2023), quite high in comparison to other applications of BERT in similar contexts (Lukmanjaya et al., 2026). While the studies covered in Lukmanjaya et al. (2026) proved the potential of NLP tools in processing varied police data, none made use of CRB finding letters, leaving them an untapped sea of information on one of the most popular forms of external police regulation. This project fills that crucial gap in the literature by manipulating the textual content of Albuquerque's CPOA into case-wise data for analysis.

Civilian oversight agencies tend to release both aggregated periodic reports on their productivity and individual case-related documentation such as finding letters. While the quarterly, semi-annual, or annual reports illustrate the agencies' productivity using descriptive statistics, they fail to qualify the nature of cases under review, and aggregation prohibits researchers from identifying case-specific trends. In contrast, the qualitative data in finding letters include rich contextual information alongside the findings of the CRB, such as descriptions of allegations under review, the timing and efficiency of investigations, and police officers involved. Developing a project for extracting and quantifying meaningful variables from this data would enable a much deeper understanding of the role and enacted purpose of the CRB, which we might then compare to its stated organizational aims to craft recommendations for improvement.

3 The Context of Albuquerque

The necessary organizations, data, and context for such a project are all present in the case of Albuquerque, New Mexico.

3.1 Albuquerque Civilian Police Oversight

The DoJ lawsuit against APD

In November 2012, the United States Department of Justice (DoJ) launched an investigation into the Albuquerque Police Department (APD) (U.S. Department of Justice, 2012). Having found in April of 2014 that the department engaged in a pattern of unreasonable use of excessive force, the DoJ entered into a consent decree (CD) with the City of Albuquerque (U.S. Department of Justice, 2014). This settlement, published in November of 2014, required a slew of reforms with the aims of "ensur[ing] police integrity, protect[ing] officer safety, and prevent[ing] the use of excessive force," (U.S. Department of Justice, 2014, p. 5). Among these reforms was the dissolution of the previous internal review board (the "Internal Review Office" or "IRO") and its replacement with an external, citizen-seated review board called the Civilian Police Oversight Agency (CPOA). The APD and CPOA operated under DoJ oversight until May of 2025 when a U.S. District Judge officially dismissed the settlement, the City of Albuquerque having reached the stipulated benchmarks of compliance (City of Albuquerque, 2025).

The CPOA and its Finding Letters

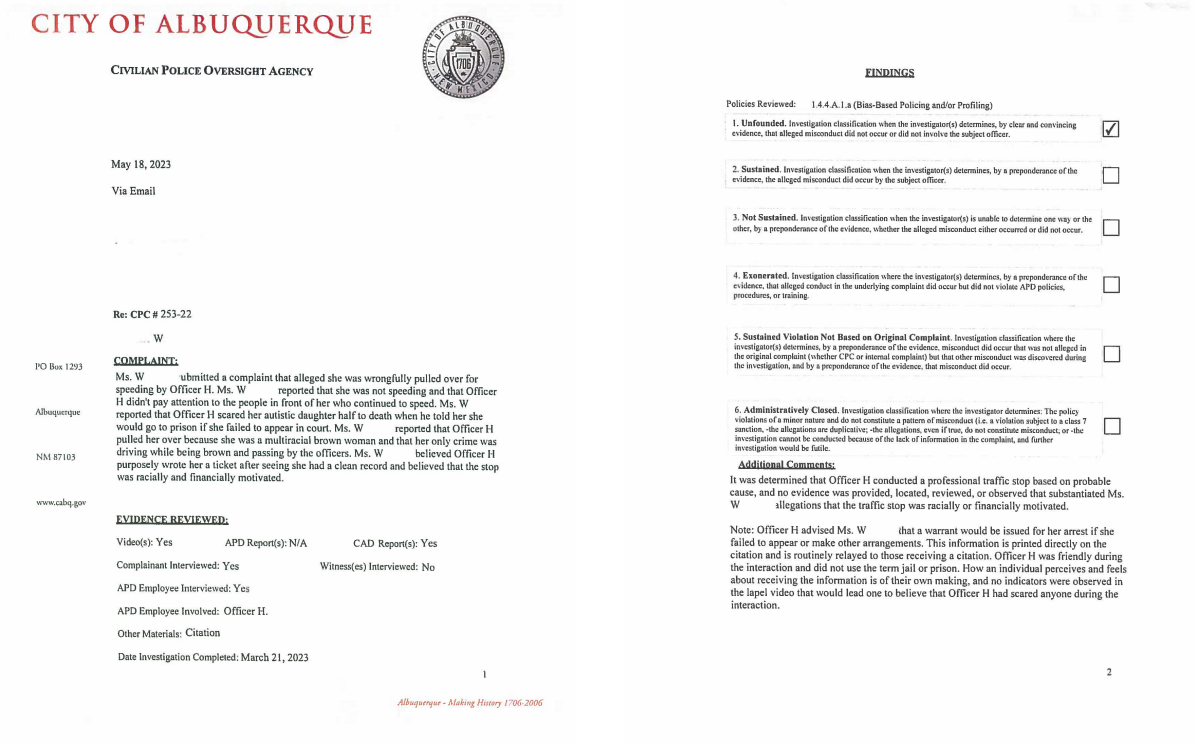
The CPOA is a CRB with government-appointed citizen board members, independent review and investigatory powers, and the discretion to recommend disciplinary action against employees of APD (City of Albuquerque, 2023). The CPOA comprises an Executive Director, five city council-appointed board members and an administrative/investigatory staff. The executive director ensures all administrative and investigative duties of the agency are completed, and sends finding letters to complainants and to APD leadership. The board (and director) are required to be at least three years

removed from employment at APD, or one year removed from working for another law enforcement agency. They are also required to learn about APD procedures (including the attendance of more than 25 hours training and two APD ride-along sessions), civil rights, and local laws and mandates (City of Albuquerque, 2023). The governing documents also recommend trainings on firearm simulations and cultural sensitivity, but these are not required (City of Albuquerque, 2023). The administrative staff (whose education and training requirements are not stated in the CPOA governing documents) conduct investigations, compiled into reports by the director, which are then reviewed by the board. While the investigatory team are unbiased in theory, they rely on APD and the Internal Affairs Bureau (IAB—the APD-internal investigatory body for potential violations of protocol or for complaints raised by APD employees) for the "reasonable" provision of most investigatory materials like on-body recording device (OBRD) footage, police reports, etc (City of Albuquerque, 2023). The CPOA, upon conducting their independent investigations, produces findings which are reported to APD's Chief of Police, who has 30 days to respond with explanations of nonconcurrence in the event that they disagree with the findings or do not enforce the disciplinary recommendations (City of Albuquerque, 2023). The CPOA is required to report periodically (at least semi-annually) to the office of the mayor, with whom they may also raise concerns over APD nonconcurrence (City of Albuquerque, 2023).

The CPOA is charged with overseeing three main types of cases: officer-involved shootings, serious incidents of use of force (UOF), and civilian complaints not deemed a part of either former category (City of Albuquerque, n.d.). Officer-involved shootings and serious UOF occurrences are also independently assessed by the IAB, and the CPOA's findings are sent directly to police leadership. In contrast, civilian complaint cases are often only considered by the CPOA, and the findings are sent directly to the complainant(s) as well as police leadership. In this way, the CPOA controls a major channel of communication between public perspectives and leaders in law enforcement. They thus maintain the power to either oppress or lift up the populace's perspectives and experiences with APD, and the bulk of their substantive work in this role is recorded in their finding letters.

The CPOA's civilian complaint finding letters, the richest source of free-text data directly connecting Albuquerque's citizens to systems of police constraint, comprise the main dataset for this thesis. Complaints may come directly from contact by a complainant or from complaints described by employees of the city or state government, as all civil servants are required to forward complaints against APD to the CPOA (City of Albuquerque, 2023). These complaints, yearly numbering between 108 (in 2020) and 306 (2022), can center on topics ranging from officer rudeness to fraudulent reporting of overtime hours to incidents of workplace sexual harassment (see Figure 3.1 for an example complaint and its findings).

Figure 3.1: An example first two pages of a finding letter from 2023



CITY OF ALBUQUERQUE

CIVILIAN POLICE OVERSIGHT AGENCY

May 18, 2023

Via Email

Re: CPC # 253-22

W

PO Box 1293

Albuquerque

NM 87103

www.cabq.gov

COMPLAINT:

Ms. W ubmitted a complaint that alleged she was wrongfully pulled over for speeding by Officer H. Ms. W reported that she was not speeding and that Officer H didn't pay attention to the people in front of her who continued to speed. Ms. W reported that Officer H scared her autistic daughter half to death when he told her she would go to prison if she failed to appear in court. Ms. W reported that Officer H pulled her over because she was a multiracial brown woman and that her only crime was driving while being brown and passing by the officers. Ms. W believed Officer H purposely wrote her a ticket after seeing she had a clean record and believed that the stop was racially and financially motivated.

EVIDENCE REVIEWED:

Video(s): Yes APD Report(s): N/A CAD Report(s): Yes

Complainant Interviewed: Yes Witness(es) Interviewed: No

APD Employee Interviewed: Yes

APD Employee Involved: Officer H.

Other Materials: Citation

Date Investigation Completed: March 21, 2023

1

Albuquerque - Making History 1706-2006

FINDINGS

Policies Reviewed: 1.4.4.A.1.a (Bias-Based Policing and/or Profiling)

1. **Unfounded**. Investigation classification when the investigator(s) determines, by clear and convincing evidence, that alleged misconduct did not occur or did not involve the subject officer. ☑

2. **Sustained**. Investigation classification when the investigator(s) determines, by a preponderance of the evidence, the alleged misconduct did occur by the subject officer. ☐

3. **Not Sustained**. Investigation classification when the investigator(s) is unable to determine one way or the other, by a preponderance of the evidence, whether the alleged misconduct either occurred or did not occur. ☐

4. **Exonerated**. Investigation classification where the investigator(s) determines, by a preponderance of the evidence, that alleged conduct in the underlying complaint did occur but did not violate APD policies, procedures, or training. ☐

5. **Sustained Violation Not Based on Original Complaint**. Investigation classification where the investigator(s) determines, by a preponderance of the evidence, misconduct did occur that was not alleged in the original complaint (whether CPC or internal complaint) but that other misconduct was discovered during the investigation, and by a preponderance of the evidence, that misconduct did occur. ☐

6. **Administratively Closed**. Investigation classification where the investigator determines: The policy violations of a minor nature and do not constitute a pattern of misconduct (i.e. a violation subject to a class 7 sanction, -the allegations are duplicative; -the allegations, even if true, do not constitute misconduct; or -the investigation cannot be conducted because of the lack of information in the complaint, and further investigation would be futile. ☐

Additional Comments:

It was determined that Officer H conducted a professional traffic stop based on probable cause, and no evidence was provided, located, reviewed, or observed that substantiated Ms. W allegations that the traffic stop was racially or financially motivated.

Note: Officer H advised Ms. W that a warrant would be issued for her arrest if she failed to appear or make other arrangements. This information is printed directly on the citation and is routinely relayed to those receiving a citation. Officer H was friendly during the interaction and did not use the term jail or prison. How an individual perceives and feels about receiving the information is of their own making, and no indicators were observed in the lapel video that would lead one to believe that Officer H had scared anyone during the interaction.

2

Upon receiving a complaint, the CPOA contacts complainants and conducts an investigation of police materials where applicable. Their findings are summarized in letters sent to complainants and, post-censorship, published online at cabq.gov along with the Chief of Police's letters of nonconcurrence. Complainants are offered the opportunity to request an appeal of CPOA findings, which is either denied or heard and summarized in finding letter documents published with the complaint finding letters. Table 3.1 displays the possible outcomes of CPOA investigations and their criteria. In earlier years, there existed a seventh option of "mediation," but this outcome was phased out for its overlap with other outcome categories.

The CPOA touts its structural independence from city government and police organizations as proof of its unbiased nature. Knowing human institutions cannot avoid bias altogether, this project seeks to explore the cases seen by the CPOA and identify trends in their outcomes. Depending on what this project reveals, the CPOA may show itself to be a pro-police organization with few recommendations for disciplinary action, a board of vindictive anti-police citizenry demanding harsh disciplinary action for any minor violation of code, or anywhere in between. The analysis of the Chief of Police's nonconcurrence adjacent to the main dataset will also inform us as to the practical impact of CPOA behaviors—if, for example, a vast majority of cases with recommendations for discipline are matched with nonconcurrence reports of lower-stakes or exonerated findings by APD leadership, it may be argued that the CPOA fills *no* substantive role besides providing potential catharsis for civilian complainants.

Table 3.1: CPOA Finding Outcomes

Outcome	Description
Unfounded	... the investigator(s) determines that alleged misconduct did not occur or did not involve the subject officer.
Sustained	... the investigator(s) determines the alleged misconduct did occur by the subject officer.
Not Sustained	... the investigator(s) is unable to determine one way or the other whether the alleged misconduct either occurred or did not occur.
Exonerated	... the investigator(s) determines that alleged conduct in the underlying complaint did occur but did not violate APD policies, procedures, or training.
Sustained Violation Not Based on Original Complaint	... the investigator(s) determines that misconduct did occur that was not alleged in the original complaint (whether CPC or internal complaint) but that other misconduct was discovered during the investigation, and that misconduct did occur
Administratively Closed	... the investigator determines: -the policy violations of a minor nature and do not constitute a pattern of misconduct; -the allegations are duplicative; -the allegations, even if true, do not constitute misconduct; or -the investigation cannot be conducted because of the lack of information in the complaint, and further investigation would be futile.

3.2 Central Inquiry

Toward the aims described above, this exploratory project examines finding letters published for the years 2018-2025 (N = 1902) and evaluates the potential for NLP tools' utility in such an application as CRB assessment. The inquiry centers on two main research questions, each with supporting/followup queries:

- RQ 1: Can supervised machine learning models stably predict the content and context of finding letters such that they are well-operationalized for analysis?

  RQ 1.1: Does OCR reliably extract pertinent information for analysis of these documents?

  RQ 1.2: Does TF-IDF outperform more complex algorithms like BERT and SBERT in text classification pipelines predicting descriptive factors of complaints?

- RQ 2: What attributes of civilian complaints against police result in disciplinary action recommendations by the CPOA? In nonconcurrence with the recommended discipline by APD leadership?

  RQ 2.1: What is the distribution of cases belonging to the variables of interest over time?

  RQ 2.2: Do complaints alleging bias-based policing correlate significantly with a higher probability of the CPOA recommending discipline?

  RQ 2.3: Does the mention of a complainant's mental health condition correlate significantly with a lower probability of the CPOA recommending discipline?

Strengths and Limitations of ABQ as Case

Focusing my inquiry on the data provided by this sole CRB has advantages and disadvantages. First, the consent decree (CD) levied by the DoJ could improve the quality of CPOA and APD data reporting, as the federal oversight between 2014 and 2025 hypothetically incentivizes greater transparency and organization. However, as a result of the institutional changes demanded by the CD, both the CPOA and APD steadily transformed during the period of DoJ review until meeting the requirements for settlement. This means necessarily that there are gradual changes to be controlled for as "noise" introduced into the dataset. Controlling for a time trend helps somewhat, but countless minor adjustments in the CPOA investigatory process would be expected yet impossible to control for individually. Still, in comparison to studies endeavoring to compare materials across police departments or even across states, the CPOA data is relatively less noisy given that there are no inter-organizational reporting differences to contend with. This single-organization focus naturally limits generalizability, but this project's exploratory aims and pioneering effort in qualifying the nature of civilian complaint texts processed by a CRB justify the tradeoff.

4 | Methodology

In the following section, I will detail the methodologies employed at each step of this project and their performance metrics. I will begin by discussing the tools I used throughout the project, moving to text extraction and preprocessing, parameter isolation, and then my process for hand-coding a random sample. Then, I will explain the tests I ran on varied potential vectorizer-regressor pipelines for classifying the data, tuning the hyperparameters on the selected "best" vectorizer-regressor classification pipeline, and applying the trained model on the full corpus of data. I will conclude this section with a discussion on my secondary dataset of information extracted from APD's nonconcurrence letters.

4.1 The Dataset

The dataset consisted of all complaint, investigation and findings content available in the finding letters published on the CPOA's website. All of the finding letters published in English and in complete format before February 20th 2026 were downloaded. A portion of early "finding letters" took more the structure of paperwork, with little unique text for analysis, and were thus omitted from the dataset.

Tools and Working Environments

The bulk of data processing (using numpy (2.1.3), scikit-learn (1.6.1), pandas (2.2.3) and statsmodels (0.14.4)) (Harris et al., 2020; Pedregosa et al., 2011; McKinney, 2010; Seabold and Perktold, 2010) and model training (using sentence-transformers (5.6.0)) (Wolf et al., 2020; Reimers and Gurevych, 2019) was completed using Python (3.13.5) (Python Software Foundation, 2026) in Jupyter Notebook (Project Jupyter, 2026). Training BERT Word Embeddings (using torch (2.12.1), transformers (5.12.1), datasets (5.0.1) and accelerate (1.14.0)) (Wolf et al., 2020) was completed in Google Colaboratory (Google, 2026).

There are a variety of algorithms that might be effective for this project, and I chose to compare the

performance of term frequency-inverse document frequency vectorization (TF-IDF), word embeddings generated by the pre-trained Bidirectional Encoder Representations from Transformers algorithm (BERT), and Sentence-BERT (SBERT) sentence embeddings. TF-IDF vectorization paired with a Naive Bayes (NB) classifier provides a strong baseline for performance (Jurafsky and Martin, 2025). I omitted the process of stemming in this context given concerns of over-conflation of semantically-like terms that I sought to differentiate, but set `df_max` to 0.8 so that the most frequent terms were excluded from the document feature matrix (Grimmer et al., 2022). BERT word embeddings had potential as an effective classification tool in this project given their ability to introduce sentence-context in weightings, and in light of their use in other police science articles like that of Hodgkinson et al. (2023). I initialized BERT with its pre-trained language representations plus a classification layer from my hand-coded data, and trained the model using a class-weighted cross-entropy loss that assigned greater weight to the minority class. Finally, SBERT, an adaptation of BERT, was a reasonable third algorithm for testing in this context because of its optimization in representing entire sentences, which might make it better for encoding the short complaint content in comparison to n-gram level embeddings (Reimers and Gurevych, 2019). Further, its computational cost is much lighter, which, given the time constraints of this project, allows for more experimentation with hyperparameter tuning.

Extraction and OCR Validation

Because text layers existed inconsistently for the letters, Tesseract OCR (5.5.2) (Tesseract OCR, 2026) was applied to the entire corpus of PDFs. Tesseract's textual output for each monthly-collection PDF needed to be parsed into unique letters, which I chose to approach with regular expressions. With much trial and error, I developed patterns for reliably extracting the content depending on the basic structure of the letter. Chunks of text that didn't get captured were systematically reviewed to ensure no finding letters were omitted.

A random stratified sample was drawn from the dataset, with five cases from each year of letter publication to ensure temporal coverage. Comprising a total of 158,655 characters (excluding spaces and after isolating for main textual content comparison) and 30,524 words, my hand-typed gold-standard sample was compared against the OCR text to test for accuracy. The results, displayed in Table 4.1, answer RQ 1.1 (*Does OCR reliably extract pertinent information for analysis of these documents?*) affirmatively. While there's no exact baseline for OCR performance in this specific context, Tesseract's performance on typed documents often produces Character Error Rates (CERs) and Word Error Rates (WERs) under 2% (Sporici et al., 2020). While this dataset had a median CER of 1.98% and WER of 4.47%, it is reasonably explained by messy censorship leaving pieces of letters to be picked up as OCR artifacts. PDF pre-processing like cropping such that text in the letter margins or boxes

Table 4.1: OCR Character and Word Error Rates

Metric	CER	WER
Mean	3.28%	5.76%
Median	1.98%	4.47%
Corpus	3.10%	5.61%

Note: Mean and Median here are casewise statistics, corpus referring to all of the text in the sample

encasing the list of CPOA findings do not interfere with the OCR could improve CER and WER. Among the finding letters, three main structural patterns emerged, and I developed strategies for cleaning boilerplate terminology and extracting case numbers, dates of interest, and CPOA findings for each one. Notable lapses in censorship of the recipient/complainant identification data, including names and addresses, were scrubbed. The differing format of the letters led me to combine "investigation" content in some cases into the “complaint” section of the letter and in other cases into the “findings” section of the letter, depending on the descriptive section header. Where there was ambiguity, "investigation" material was prepended to the “findings” material.

4.2 The Variables

Directly Extracted Variables

I utilized regular expressions to extract several contextual parameters. The simplest to extract were `case_number`, `letter_date`, `officer_involved` (in the later formats where it is directly reported), and `date_investigation_complete` (also in later formats). More complex were the complaint and incident dates, which relied on a few potential textual anchors, and, even with multiple rounds of verification and adjustments in the regular expressions, maintained some missing captures. Contributing further to missingness, many finding letters never explicitly cited a complaint or incident date. For letters citing multiple incident dates, extraction captured the earliest date mentioned positionally (usually the earliest temporally as well). Attempts to capture the location of events were unreliable due to underreporting of exact location and/or censorship of addresses to protect complainants’ personal information. Findings were extracted with a series of regular expression capture patterns, working recursively such that the most reliable pattern for capturing across the dataset was used first, then missingness was increasingly filled in with additional capture patterns. To ensure the validity of my extractions, I compared the information to that of the hand-coded sample, resulting in the following metrics of deviation:

Table 4.2: Reg-Ex extracted variables vs hand-coded gold sample

Variable	Mean Difference	Std Dev	Max Abs Diff	Diff Pct Missing
`case_number`	0	0	0	0.00%
`letter_date`	0.000	0.000	0	0.00%
`other_finding`	-0.005	0.22	1	0.00%
`incident_date`	3.949	34.53	305	18.94%
`complaint_date`	0.000	0.000	0	2.10%
`date_inves_complete`	0.000	0.000	0	0.00%

Note: Date differences measured in days
Note: 4 out of 190 rows had differences in incident dates, whereas 9 differed regarding CPOA findings

As anticipated, the extracted incident date missingness was higher when compared to the hand-coded sample. All efforts to bridge that gap in missingness, however, resulted in even greater capture inaccuracies. As such, I utilized `letter_date` (which had zero missingness after a few manual corrections) to order my data chronologically.

Duplicativeness was an additional challenge in maintaining consistency throughout the data. Duplicative publication of the exact same finding letters was easily collapsed, but a problem emerged from the changing letter publication regulations over the years. In early years, nearly all case numbers corresponded to a single finding letter addressed to a sole recipient and containing all allegations against all relevant employees of APD. Beginning around the end of 2021, we begin to see an increase in the letters produced by the CPOA in great part because they start publishing separate letters addressed to the various different complainants making the same complaint against the same officers *and* separate letters for each distinct member of APD cited in the complaint. With coding assistance from OpenAI's ChatGPT (OpenAI, 2026), I created an algorithm for combining all letters with entirely duplicative textual content into one observation, with the combined findings collated in the findings (`f_1`, `f_2`, etc.) columns, such that all findings were maintained for each unique complaint raised. This does not mean that the resulting dataset comprised one observation per case number, but one observation per unique complaint and findings *textual content*, which supported consistency in information reporting across the timeframe of the dataset. This process, along with the removal of appeal-denial letters (essentially identical in content and non-informative to the particulars of any case), narrowed the dataset by 95 observations, bringing the total to N = 1902.

Hand-Coded Variables

I designed my hand-coded variables meant for training the machine learning models in a manner aimed at reducing ambiguity or "gray area" as much as possible. Many of the studies utilizing NLP tools in police science target sentiment analysis tasks using large language models (LLMs) for the purpose of creating structure (Sarzaeim et al., 2023), but this project is focused on testing less complex, open-use algorithms. In the context of honing term frequency-inverse document frequency, word

embedding, or sentence-vectorization models, maintaining as clear a difference as possible between the "positive" and "negative" classes is essential (Grimmer et al., 2022). Thus, in contrast to other studies attempting ordinal scales to describe phenomena such as officer demeanor (Dlugolenski et al., 2024), I developed binary coding criteria for all parameters of interest to be as clear-cut as possible. The distinction between a "0" (not present) and "1" (present) for each term was refined such that substantive meaning was retained (e.g. an officer's handling of civilians' property) without narrowing the focus to the extent that each classifier centered on too specific a phenomenon (e.g. property being missing but not destroyed).

Because CRB unstructured textual data has yet to be explored in the literature, the thematic choice in variables of interest relied on both the concepts common to law enforcement oversight literature and the prevalent themes in this particular dataset.

Table 4.3: Hand-Coding Criteria and Counts

Variable	Criteria	N (1)	N (0)	Included?
`bias_complaint`	Explicit mention of discrimination or biased treatment on the basis of identity	11	179	Yes
`weapon_complaint`	Allegation of officer pointing a weapon at (or using a weapon on) a civilian	5	185	No
`demeanor_complaint`	Complaint mentions rudeness, profanity, or unprofessionalism in an officer's demeanor	70	120	Yes
`regulatory_complaint`	Complaint references an officer incorrectly recording information relating to an incident, being slow in filing reports, not using an OBRD, or refusing to provide badge numbers/identification	36	154	Yes
`property_complaint`	Complaint mentions missing or destroyed property, e.g. alleging officers failed to secure or return property	14	176	No
`mental_health`	Complaint or findings mention a mental health condition of the complainant or their being psychiatrically evaluated	10	180	Yes
`traffic_scenario`	Indicator that complaint resulted from: a traffic stop (excluding cases where officers run plates and find an active warrant or stolen car), car crash (between complainant and another individual, officers included), or a citation relating to either	34	156	Yes
`jurisdiction_flag`	Indicator of whether complaint relates to issue outside of APD jurisdiction	16	174	No
`duplicative_flag`	Indicator of whether complaint was closed because of its duplicative nature with another complaint	2	188	No
`withdrawal_flag`	Indicator of whether complaint was closed because of its being withdrawn by the complainant	10	180	No
`other_finding`	Indicator that at least one of the findings is: sustained, sustained not based on original complaint, or mediation	34	156	Yes

Note: "OBRD" refers to On-Body-Recording-Device

Content Variables

Parameters relating to the allegations made in the complaint are `bias_complaint`, `weapon_complaint`, `demeanor_complaint`, `regulatory_complaint`, and `property_complaint`. Many studies focus on biased-based policing (Parks and Kirby, 2021; De Orio, 2022) and UOF incidents (Rogers et al., 2019),

inspiring my development of the `bias` and `weapon` variables. Unfortunately, because there were so few positive cases of `weapon_complaint` in my hand-coded sample (due to the CPOA's general handling of officer-involved shootings separately from civilian complaints), I excluded it from my analysis. I also wanted to capture, via `demeanor_complaint` and `regulatory_complaint`, the common themes of complaints about an officer's demeanor (e.g. "the officer was dismissive and rude") or reporting errors (e.g. "the officer lied in the crash report"). Finally, I noticed a theme of complainants' concern over the handling of their property (e.g. "the officer took my purse and never returned it"), which I sought to capture with `property_complaint`. Challenges in reliable cross-validated predictions with any algorithm prohibited my use of this variable in my final analysis, which I suspect was due to the varied and often ambiguous labeling of a complainant's property—it could pertain to any physical object, from "jacket" to "wallet" to "gun," the semantic spread of which would naturally challenge convergence.

Context Variables

My two contextual variables that inform the setting and framing of the complaints are `mental_health` and `traffic_scenario`. While police literature might focus on different rates of force or arrest applied to those living with mental illnesses (Rogers et al., 2019), in the context of civilian review, this parameter might be most informative of the oversight board's own biases. If `mental_health` is predictive of cases against officers being dismissed, it could indicate the review board members viewing mentally ill complainants as less credible than those whose conditions are not explicitly mentioned. I also observed that many complaints surrounded the officers' handling of events just after a car accident or a traffic stop. Given its thematic prevalence in the dataset and literature alike (Parks and Kirby, 2021; Dlugolenski et al., 2024), it felt informative to include the `traffic_scenario` parameter.

Validation

I employed a few different strategies to keep my hand-coding process as methodologically sound as possible despite being the sole contributor to this project. I used scikit-learn's StratifiedShuffleSplit to draw a random sample of 10% of the corpus (N = 190) stratified across years of letter publication for hand-coding (Pedregosa et al., 2011). The yearly stratification was crucial given the change in the data format and shift in content over time. In place of inter-coder reliability tests with another human coder or with a cost-protected model like Atlas.TI, I chose to utilize the open-source SBERT sentence encoder to identify cases where semantic similarity did not align with my coding choices (Reimers and Gurevych, 2019). I then evaluated each flagged observation and resolved inconsistencies where sensible, changing a total of four coding decisions.

4.3 Classifiers

Because this project sought to compare supervised machine learning models in CRB free-text analysis, three main text vectorizers (TF-IDF, BERT, SBERT) were considered and assessed for use, with SBERT consistently outperforming the others.

Initial Testing

TF-IDF, BERT, and SBERT text representations were first generated from a stratified train-test split (80% train, 20% test set) of the hand-coded data. TF-IDF and SBERT representations were used to train a variety of classifiers (NB, Logistic Regression, Support Vector Machine, Random Forest), and BERT representations were trained with class-weighted loss equations. I compared the training on either strictly the complaint text or the complaint and findings text together for each variable. This was done with the consideration that some factors might be best captured in the complaint alone, with the findings text adding only noise, whereas others might rely on the text of findings (for example, the `mental_health` variable criteria necessitates training on the entire letter text). My BERT algorithm performed similarly to the TF-IDF/Naive Bayes baseline, and was so computationally costly, that I moved forward with honing on TF-IDF and SBERT alone. Performance varied across TF-IDF/SBERT vectorizer-classifier combinations. Given the small number of positive cases in some parameters, I performed 5-fold stratified cross-validation to train over different portions of the training data to provide more robust estimates of out-of-sample accuracy. Over most metrics, SBERT consistently outperformed TF-IDF models, except in the case of the `regulatory_pred` variable, in which the TF-IDF-LogisticRegression pipeline performed slightly better. This provides a generally negative answer to RQ 1.2 (*Does TF-IDF outperform more complex algorithms like BERT and SBERT in text classification pipelines predicting descriptive factors of complaints?*). Certain regressors managed the case imbalance better than others, but LinearSVC and LogisticRegression often performed comparably.

Tuning Hyperparameters

For each best-performing combination of regressors with either TF-IDF or SBERT, I ran hyperparameter tuning using GridSearchCV over a grid of parameter options, optimizing for maximizing the Macro F1 score. I did the same for the SBERT−LinearSVC pipeline on every parameter, as that was the most common high-performing pair. The SBERT-SVC pairing with tuned hyperparameters performed best overall for achieving consistent, conservative accuracy in both majority and minority classes across variables. Optimizing based on the Macro F1 score was intended to avoid overly optimistic performance estimates from the class-balanced models. The final models

are summarized in Table 4.4.

Table 4.4: Final Classifier Model Specifications

Variable	C	Loss	CV Macro F1	Test-Set Accuracy	Test-Set F1
`traffic_scenario`	1	Hinge	0.78	0.89	0.71
`mental_health`	1	Hinge	0.70	0.92	0.40
`demeanor_complaint`	10	Hinge	0.64	0.68	0.60
`regulatory_complaint`	10	Hinge	0.65	0.89	0.67
`bias_complaint`	1	Squared Hinge	0.64	0.97	0.67

Note: All observations reflect results of SBERT sentence embeddings with LinearSVC tested over a parameter grid to converge on hyperparameter settings maximizing Macro F1 scores. "C" shows the hyperparameter setting for balancing costs of classification errors and adjustments in the decision boundary margin size while "Loss" shows the hyperparameter setting for adjusting the degree to which margin violations are penalized. All models utilized balanced class weights.

Returning to my first overarching question of interest:

RQ 1: Can supervised machine learning models stably predict the content and context of finding letters such that they are well-operationalized for analysis?

The answer differs based on the variable. All classifiers had moderate success, barring `property_complaint`, which was nixed from final analysis for the model's inability to optimize on Macro F1 without tanking F1 scores down to a mere 0.2. The remaining content variables performed moderately, with cross-validated Macro F1 scores ranging between 0.64 and 0.78. While these scores are not particularly strong compared to some in a review of studies on NLP uses in police science (Lukmanjaya et al., 2026), they are significant advancements from the TF-IDF baseline and class-balanced Dummy Classifiers in terms of validity and performance. Recommendations for future research improving these outcomes are outlined in the concluding chapter of this thesis.
Applied to the full corpus, the models produced conservative predictions of each factor's presence in the dataset, summarized in Table 4.5.

Table 4.5: Total predicted positive/negative counts

Variable	N (1)	N (0)
`traffic_pred`	496	1406
`mh_pred`	73	1829
`demeanor_pred`	733	1169
`regulatory_pred`	298	1604
`bias_pred`	106	1796
`other_finding`	394	1508
`nonconcurrence_flag`	70	1832
N	1902	

Note: "other_finding" refers to at least one finding captured with regular expressions being "sustained"

4.4 Nonconcurrence Dataset

In addition to the main body of data, I collected a second dataset of nonconcurrence cases to assess the instances in which APD leadership altered or ignored the disciplinary action recommended by CPOA. Officially, the Chief of Police is required to write letters in all instances of nonconcurrence (City of Albuquerque, 2023), but given the number of minor infractions and recommendations for "verbal reprimands," which are easily ignored without documentation, I have some doubts that the totality of nonconcurrence cases is reported. Of those publically accessible, though, I hand-collected information about the cases (see examples in Table 4.6). There was only one instance in which APD applied stricter disciplinary action than CPOA, and in an effort to maintain a consistent concept of "nonconcurrence" referring to APD imposing lighter disciplinary action than CPOA, the case was omitted. Cases where decision and disciplinary differences were ambiguous or insignificant (e.g. changing from "not sustained" to "unfounded", N = 6) were also omitted.

Table 4.6: Three cases from the nonconcurrence dataset

Case Number	Employee	CPOA Recommendation	Discipline Taken	Officer Gender
192-22	Officer H	Written Reprimand	None	male
149-21	Officer C	Verbal Reprimand	NDCA	unknown
149-21	Officer C-H	8-hour suspension	Written Reprimand	female
101-25	Officer Q	48-hour suspension	None	male

This data was matched to the corpus of finding letter data based on case number, and a new binary parameter (`nonconcurrence_flag`) was added to all observations indicating whether the case had at least one CPOA recommendation of disciplinary action modified (in levity) by APD.

5 | Results and Discussion

Post-application of my predictive models, variable label formatting changed from `var_complaint` to `var_pred` and I added a `nothing_pred` term for observations that had none of my variables of interest predicted as present. I interpret the following statistical analyses with great caution given the moderate performance of the classification models.

5.1 Association Tests

An important step on the way to inference is measuring association among my parameters of interest. A Chi-Squared (χ^2) test showed that a few significant associations existed among the variables, summarized in Table 5.1.

The strongest association between complaints with `regulatory` content in the `traffic_scenario` context makes perfect sense. Because Americans rely on accurate, prompt police reports to make successful insurance claims, many complaints regarding the timeliness or accuracy of police reporting relate to car accidents. Further, Americans refuting a traffic citation might be complaining that officers misreported details of their behavior behind the wheel. The second strongest association, `bias-traffic` complaints, could indicate a number of phenomena. Sociological and police science literature cite traffic stops as a mechanism for discriminatory surveillance of non-white residents in the U.S. (Alexander, 2010), which might be an underlying factor in this association. Still, given the positive-class imbalance between bias complaints and traffic scenarios, I am reluctant to view the association as precipitated by anything more than chance. Other associations never rise above a Phi

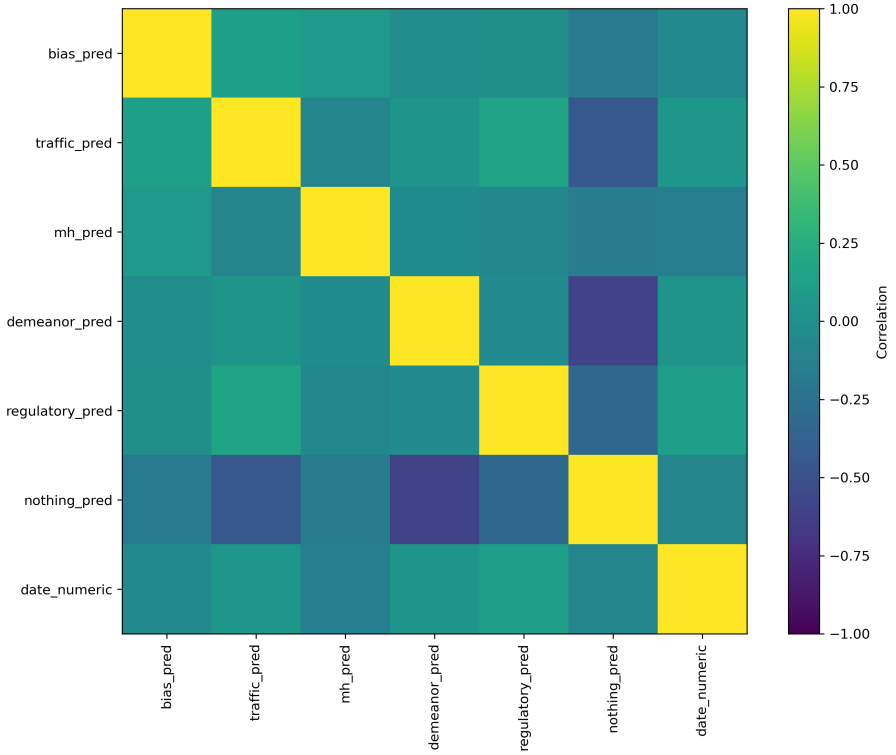
Table 5.1: Top χ^2 associations among variables.

Variable 1	Variable 2	chi2	p-value	p-adjusted	phi
`regulatory_pred`	`traffic_pred`	53.24	2.95 e-13	2.95 e-12	0.17
`bias_pred`	`traffic_pred`	29.50	5.60 e-08	2.80 e-07	0.12
`mh_pred`	`traffic_pred`	15.62	7.76 e-05	2.60 e-04	0.09
`mh_pred`	`bias_pred`	11.20	8.19 e-04	2.05 e-03	0.08
`regulatory_pred`	`mh_pred`	6.79	9.15 e-03	1.83 e-02	0.06

coefficient of 0.1, indicating that association, even if statistically significant, is unlikely to be substantively meaningful.

The correlation matrix below (Figure 5.1) shows all of the variables, including `nothing_pred`, which would necessarily be perfectly negatively associated with the sum of the my factors, and naturally presents as somewhat negatively associated with the more common variables.

Figure 5.1: Variable Correlation Matrix



5.2 Descriptive Statistics

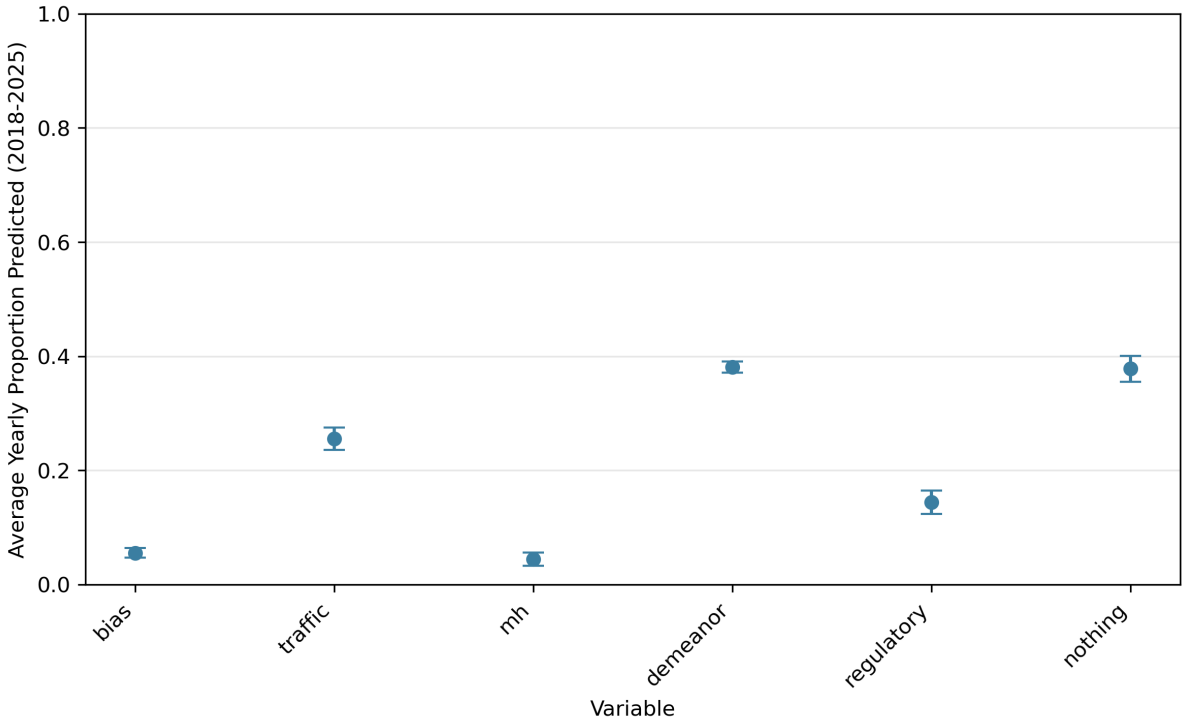
Returning to RQ 2.1 (*What is the distribution of cases belonging to the variables of interest over time?*), we see that complaint content prevalence changes over the years, with bias-based policing allegations peaking in 2018 and remaining low throughout, whereas regulatory and traffic-related complaints peak later in 2024 and 2025 respectively. A summary of mean distribution across year groups (marginal cases dated 2017 and 2026 being grouped with 2018 and 2025 respectively) is summarized in Table 5.2 below.

Table 5.2: Variable distribution over time

variable	max_prop	max_year	mean_prop	se
traffic_pred	0.33	2025	0.26	0.02
mh_pred	0.11	2018	0.04	0.01
demeanor_pred	0.42	2023	0.38	0.01
regulatory_pred	0.23	2024	0.14	0.02
bias	0.11	2018	0.06	0.01
nothing_pred	0.49	2019	0.38	0.02

My variables are present in a decent proportion of the total cases, with an average of 38% of cases across the timeframe not falling into any of them. Among content-related variables, the "demeanor" feature is most prevalent, predicted to be present in an average of 38% of cases across the years, while bias-based allegations are predicted in only about 6% of yearly cases on average. Figure 5.2 plots the average yearly prevalence of each parameter side-by-side.

Figure 5.2: Average proportion of cases belonging to each variable



Plotting the observations over time (see Figure 5.3) shows some interesting trends. The annual total of cases fluctuates between 184 in 2019 and 306 in 2022, excepting a unique low in 2020 of 108, presumably due to the COVID-19 pandemic. 2021 saw the highest proportion of nonconcurrence among cases with sustained CPOA findings (54%), while 2022 saw the highest case volume overall. Among cases where complaint dates were noted, the CPOA took an average of 158 days from the date of complaint to complete their investigations and release finding letters. The proportion of cases with

sustained findings by the CPOA generally increased across the years, beginning at a meager 4% (8 out of 223) in 2018 and increasing to 53% by the end of 2025 (147 out of 278). This extreme change over time will be kept in mind in the coming analysis.

Figure 5.3: Proportion of sustained findings and nonconcurrence over time

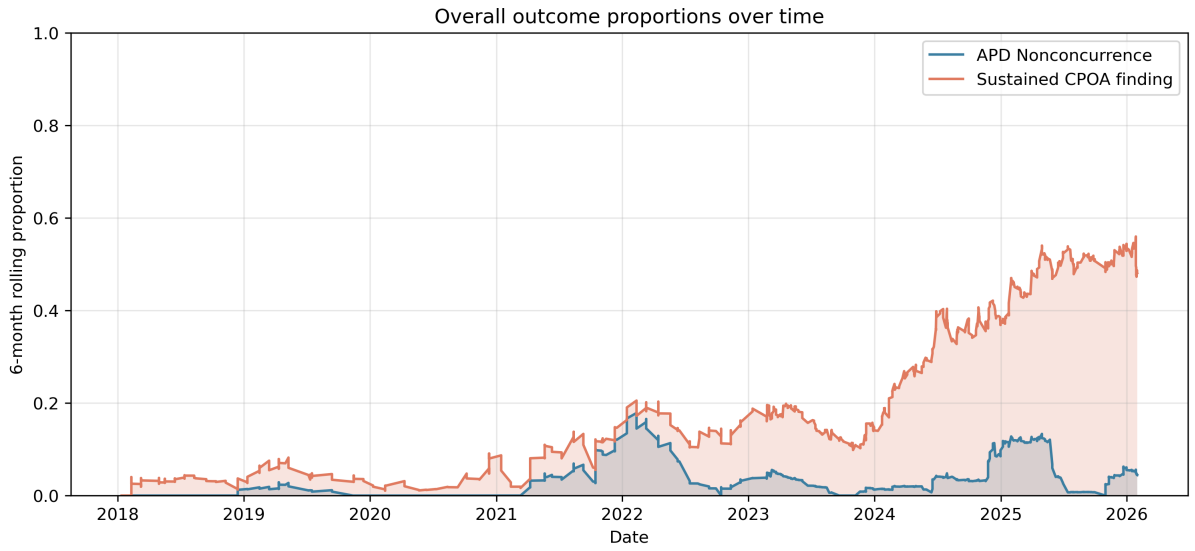


Figure 5.4 displays the proportion of sustained or nonconcurrent cases captured by my variables over time, while Figure 5.5 shows the same proportions for cases not captured by my variables. Temporally early differences between Figure 5.4 and Figure 5.5 might signify that my parameters do not do as well to represent the themes of older finding letters than more recent ones. Similarity in proportional composition of the complaints with sustained or nonconcurrent findings, however, indicates they do not capture an imbalanced section of the data on either of those metrics. This is corroborated by the non-significance of the `nothing_pred` coefficient in the final logistic regression model I ran on cases with none of my variables predicted present, described at the end of the following section.

Figure 5.4: Proportion of sustained/nonconcurrent findings captured by my variables

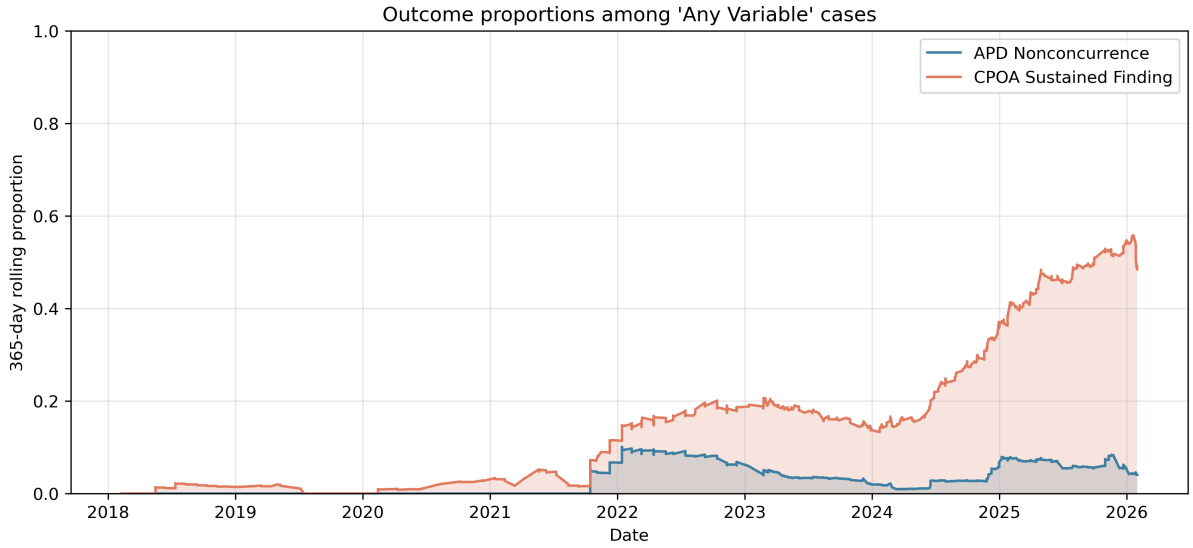
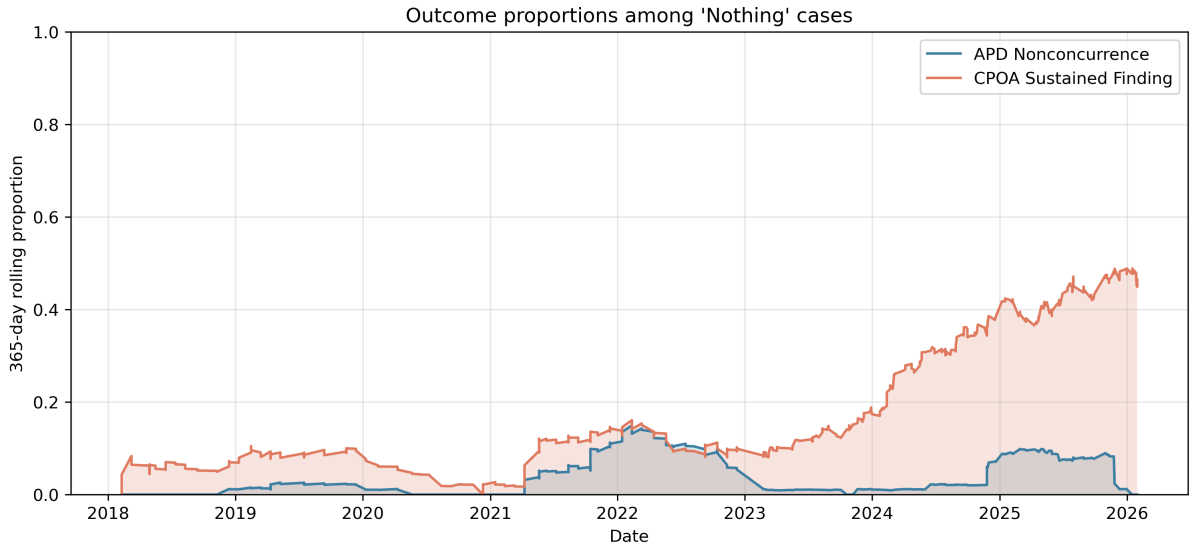


Figure 5.5: Proportion of sustained/nonconcurrent findings **not** captured by my variables



5.3 Logit Models

Discerning the relationship between my variables of choice and the dependent variable (sustained CPOA findings against officers) is possible with the application of generalized linear models (GLMs) with a logit link (or, multivariable logistic regression models). Logit-link GLMs permitted me to control for multiple parameters while restricting the outcome to probabilities ranging from 0 to 1, consistent with the structure of my data (Long, 1997). The generic formula of logistic regression is presented in Equation 1 below, where $P(Y_i = 1)$ represents the probability of sustained findings, β_0

represents the log-odds coefficient of the intercept term, and β_k represent log-odds coefficients for my input variable(s). Controlling for potential heteroskedasticity with robust standard errors resulted in minimal difference, suggesting that the conventional logistic standard error calculation is appropriate for this data. Testing for interaction between my variables was limited to few cases due to the infrequent overlap of positive classes among variables. For each term, I also constructed a 95% confidence interval around the difference in mean positive outcomes (sustained CPOA findings) associated with the presence of the term by conducting resampling with replacement for the positive and negative classes and comparing the averages. In the coming sections I will provide the results of both the overall difference in means and the logistic regressions estimating each independent variable's influence on the output term, with additive controls for contextual parameters `mental_health` and `traffic_scenario` as well as time effects (`date_numeric`).

$$\text{logit}(P(Y_i = 1)) = \beta_0 + \sum_{k=1}^{K} \beta_k X_{ki}. \tag{1}$$

Contextual Variables

The `traffic_pred` variable lacks significance in the 95% confidence interval developed through bootstrap resampling positive and negative classes (`n_iter = 10,000`) and comparing mean outcomes across the entire dataset, but presents with significance at the $p < 0.01$ level when the time control parameter (`date_numeric`) is added in the GLM. This makes sense, as traffic-related events are more prevalent in the later years as exemplified in Table 5.2. As we will see in the coming experimentations with different combinations of predictor terms, the `traffic_pred` term retains a small but consistent positive indicator across varied models.

Exponentiating the coefficients to achieve the odds ratios, we can interpret the output of these logit-link GLMs as indicating that complaints classified by the model as having a traffic-related context were, on average, associated with 45% higher odds of resulting in sustained findings by the CPOA, decreasing to 41% higher odds when controlling for the `mental_health` contextual variable.

Table 5.3: Traffic Logit Models

	(1)	(2)
Intercept	−4.16***	−4.335***
	(0.240)	(0.241)
traffic_pred	0.378***	0.344**
	(0.134)	(0.134)
date_numeric	0.001***	0.001***
	(0.000)	(0.000)
mh_pred		−2.327**
		(1.021)
N	1,902	1,902
Pseudo R^2	0.157	0.163

* p<.1, ** p<.05, ***p<.01

Note: Binary logistic regression models estimating the association between predicted `traffic_scenario` *complaints and the likelihood of a sustained CPOA finding (*`other_finding` *= 1). Model (2) adjusts for the* `mental_health` *variable. Standard errors in parentheses.*

When we isolate the predicted occurrence of a complainant's mental health being mentioned, `mh_pred`, we see a stronger, negative coefficient. This result is consistent across models including `mh_pred`, but it must be acknowledged that alongside the increase in the coefficient, we see an expansion of the standard errors. This is due to the relative rarity of the term, which left only one positive case overlapping in `mh_pred` and `other_finding`. Still, using bootstrap resampling to identify a 95% confidence interval across all years also yielded significance for this variable, the coefficient for which consistently retains significance at the $p < 0.05$ level in the GLMs. I will not interpret the odds ratios from coefficients shown here, as they are likely intensified by the small sample involved.

Table 5.4: Mental Health Logit Model

Intercept	−4.270***
	(0.240)
mh_pred	−2.421**
	(1.021)
date_numeric	0.001***
	(0.000)
N	1,902
Pseudo R^2	0.160

* p<.1, ** p<.05, ***p<.01

Note: Binary logistic regression models estimating the association between predicted `mental_health` *complaints and the likelihood of a sustained CPOA finding (*`other_finding` *= 1). Model (2) adjusts for the* `traffic_scenario` *variable. Standard errors in parentheses.*

Content Variables

Bootstrap resampling produced a confidence interval for a difference in means between the positive and negative classes of the `demeanor` variable that contained zero, not indicating significance. Adding a control for time trend in the multivariable logistic regression, we still see no significance in the

demeanor_pred feature as a predictor of case outcomes. Table 5.5 shows the coefficient outputs of a base model that includes only demeanor_pred and the date_numeric time effects term beside a those of a second model that controls additively for the contextual variables. These results prohibit the rejection of the null hypothesis that the demeanor variable is not significantly associated with sustained outcomes against APD.

Table 5.5: Demeanor Logit Models

	(1)	**(2)**
Intercept	−4.299***	−4.282***
	(0.242)	(0.245)
demeanor_pred	−0.134	−0.139
	(0.126)	(0.127)
date_numeric	0.001***	0.001***
	(0.000)	(0.000)
traffic_pred		0.345**
		(0.134)
mh_pred		−2.333**
		(1.022)
N	1,902	1,902
Pseudo R^2	0.153	0.163

* p<.1, ** p<.05, ***p<.01
Note: Binary logistic regression models estimating the association between predicted demeanor complaints and the likelihood of a sustained CPOA finding (`other_finding` = 1). Model (2) adjusts for contextual variables. Standard errors in parentheses.

Akin to what we observed in analyzing the traffic_pred variable, the regulatory_pred feature shows no significance when averaged over time in the bootstrap resampling experiment, but gains significance in the GLM with a control for time effects. Table 5.6 shows the results of the base model and the model controlling for contextual terms, with no significant change in the coefficient or standard errors. Because a large portion of the dataset belonged to both the regulatory and traffic parameters, I experimented with interaction such that I might further examine their potential overlap in capturing a phenomenon predictive of the independent variable. The interaction term, traffic_pred:regulatory_pred, held significance at the $p < 0.1$ level, while all other coefficients (besides time control) lost significance due to the increase in standard errors. The natural increase in multicollinearity across the interacting features here, in addition to a lacking improvement in Pseudo R^2 suggests that the additive model is most appropriate for interpretation. Notable anyhow is that traffic and regulatory coefficients shrink in size when combined in Model 2 as compared to either base model, corroborating the evidence of association noted in the χ^2 tests conducted earlier in this chapter.

Table 5.6: Regulatory Logit Models

	(1)	(2)
Intercept	−4.402***	−4.36***
	(0.241)	(0.243)
regulatory_pred	0.457***	0.375**
	(0.152)	(0.154)
date_numeric	0.001***	0.001***
	(0.000)	(0.000)
traffic_pred		0.288**
		(0.136)
mh_pred		−2.281**
		(1.021)
N	1,902	1,902
Pseudo R^2	0.158	0.166

* p<.1, ** p<.05, ***p<.01
Note: Binary logistic regression models estimating the association between predicted regulatory complaints and the likelihood of a sustained CPOA finding (`other_finding` = 1). Model (2) adjusts for contextual variables. Standard errors in parentheses.

The bootstrap resampled difference in means showed no significance for the `bias_pred` variable across the dataset. Adding a control for time effects with the multivariable logistic model, the `bias_pred` coefficient is strong and negative ($\beta = -0.492$) with a p-value of 0.11. After adding in the contextual confounders, `traffic_pred` and `mh_pred`, the coefficient for `bias_pred` gains significance at the $p < 0.1$ level ($p = 0.09$). The coefficient is stable in sign and strength across the models, suggesting some evidence for a negative association, but the tendency of the significance level to hover around the 0.1 threshold for significance gives me lower confidence in the `bias_pred` feature as a predictor of case outcomes. A greater sample size would lend more authority to these results.

Table 5.7: Bias Logit Models

	(1)	(2)
Intercept	−4.325***	−4.314***
	(0.239)	(0.241)
bias_pred	−0.492	−0.516*
	(0.313)	(0.315)
date_numeric	0.001***	0.001***
	(0.000)	(0.000)
traffic_pred		0.367***
		(0.135)
mh_pred		−2.268**
		(1.021)
N	1,902	1,902
Pseudo R^2	0.154	0.164

* p<.1, ** p<.05, ***p<.01
Note: Binary logistic regression models estimating the association between predicted bias complaints and the likelihood of a sustained CPOA finding (`other_finding` = 1). Model (2) adjusts for contextual variables. Standard errors in parentheses.

Nothing Variable

To support the validity of my variables, I also estimated a model with the `nothing_pred` as the input variable and the `date_numeric` term controlling for time effects. The resulting coefficient was insignificant, suggesting that the data remaining uncaptured by my parameters does not systematically predict the CPOA's findings.

6 | Conclusions and Limitations

6.1 Conclusions

This project sought to create a dataset from the free-text data found in the publically available CPOA finding letters relating to civilian complaints. Unique variables describing the content and context of complaints were developed with consideration to themes prevalent in the literature and in the complaint letters themselves, and a hand-coded sample was used to produce a body of SBERT sentence embeddings. These were used to train LinearSVC models with five-fold cross-validation optimizing for Macro F1 scores resulting in moderate accuracy for most classifiers. A separate dataset on APD leadership casewise nonconcurrence was constructed by hand and utilized alongside the main dataset for analysis. The application of GLMs with the logit link function showed contextual variables to be significant in predicting the log-odds of sustained findings, whereas most content-related parameters had little predictive power. The project as a whole serves as an important first step in investigating the prevalent themes of civilian-police relationships as mediated by the CPOA. My takeaway conclusions are outlined here, followed by my recommendations for the CPOA, the study limitations, and future research directions.

Case Volume and PSAs

The notable changes in case volume over time might be related to the expansion of the Police Service Aide (PSA) program in Albuquerque. While it could be that the CPOA's community outreach improved over time through other means, the increase in complaint volume over time co-occurs with increases in traffic-related and regulatory-related complaints. At the same time, the city of Albuquerque was expanding both membership and responsibilities of the PSA force, a group of trained individuals that assist in traffic control and responding to vehicular collisions (City of Albuquerque, 2024). Though entirely theoretical, I hypothesize that the coinciding expansion of the PSA program could have created new routes for the public to contact the CPOA, thus increasing the number of civilians voicing their concerns to the agency.

Nonconcurrence

Although the sparse presence of nonconcurrence in the dataset prohibits its analysis in relation to my variables, the results of my descriptive statistics bear mentioning. Of the nonconcurrence cases in which the employee's gender was apparent (gendered pronouns utilized), 51 cases involved male employees and 19 involved female employees. The most recent publically available gender distribution of APD officers that I could find stated that 14% of sworn APD officers in 2016 were female. The higher proportion of female officers or APD employees in the nonconcurrence dataset (27 %) does not seem particularly unusual given the likelihood of higher female presence in APD since 2016 and among PSAs and telecommunication operators. Overall, 20.72 % of the dataset included "sustained" findings against officers' misconduct and 3.68% prevalence of nonconcurrence by APD. Among cases with sustained findings, 13.71% had nonconcurrence flags, meaning that 17.88% of cases in the entire dataset resulted (hypothetically) in disciplinary action of some form. While the average data indicates a lack of APD interference with CPOA disciplinary recommendations, the yearly rate of nonconcurrence peaked in 2021 at a staggering 54%, which is disconcerting. In light of such intense fluctuation in nonconcurrence and the six-month period between the end of 2021 and the beginning of 2022 where we see a near exact overlap in sustained findings and nonconcurrence, I am reluctant to trust the transparency of APD's leadership in reporting their altered disciplinary actions to the CPOA for publication online.

Logit Results

While we could not answer the overarching RQ 2 in its entirety (*What attributes of civilian complaints against police result in disciplinary action recommendations by the CPOA? In nonconcurrence with the recommended discipline by APD leadership?*) due to the limited number of nonconcurrence cases, we did find significant predictors of CPOA "sustained" findings in the `traffic_scenario` and `regulatory_complaint` variables. Though treated with less confidence due to sample size constraints, the `mental_health` and `bias_complaint` indicators reaped significance as well. The following paragraphs will expand on these results.

Given the covariance previously noted between the `traffic_pred` and `regulatory_pred` variables, we might understand their positive coefficients in the GLMs to mean that traffic-related complaints are most likely to highlight distinct regulatory violations that can be easily checked. For example, if a complainant alleges that a report was not filed in a timely manner, CPOA investigators might check police records and time logs to see whether protocol was followed. This permits much less discretionary lenience on behalf of the CPOA, which could explain the consistent positive signals `traffic_pred` and `regulatory_pred` give for predicting sustained findings.

Coefficients for allegations of bias-based policing were consistently negative and hovered around significance at the 95% level, giving an unclear answer for RQ 2.2 (*Do complaints alleging bias-based policing correlate significantly with a higher probability of the CPOA recommending discipline?*). The possibility of `bias` allegations having this relationship with case outcomes in the CPOA suggests that the board may be prejudiced in favor of APD and reluctant to acknowledge it as part of a broader American institution with a history of racial (Alexander, 2010), gendered (Panfil, 2026), and socioeconomically (Parks and Kirby, 2021) discriminatory practices.

The `mental_health` variable being a significant predictor of sustained findings tentatively provides a positive answer to RQ 2.3 (*Does the mention of a complainant's mental health condition correlate significantly with a lower probability of the CPOA recommending discipline?*), but it could reflect a variety of phenomena. First, it bears repeating that the hand-coded sample of 190 finding letters contained only 10 cases presenting with the conditions of this parameter, mediating our interpretations of significance. Given the stigma surrounding mental health and the historic mistreatment of those with severe mental illness, we might jump to the conclusion that the mention of mental illness of the complainant is wavering investigators' sense of the complainant's credibility, and thus a higher portion of cases are closed without consequences for APD. This is one of many potential explanations, however. It could also be that complainants with mental illness are more likely to make fantastical allegations against officers that lack grounding in reality, or that complainants with mental illnesses face more barriers in communicating their complaints with enough clarity for the CPOA to investigate it fully. Regardless, the potential association between mentions of a complainant's mental condition and lower odds of sustained findings against officers warrants further investigation.

Taken in sum, the results of my analyses provide some preliminary evidence of the CPOA utilizing its discretionary powers in lenience toward APD employees. Knowing that the CPOA board (responsible for disciplinary recommendations) is in direct conversation with APD personnel much more frequently than complainants and that they are required to undergo over twenty hours of APD training prior to working on the board (City of Albuquerque, 2023), this is not entirely surprising. While there are a few places where error could leak into the structure of this project, I took careful steps to produce my data with conservative estimates so as not to bias my results. More investigation into the relationship between civilian boards like the CPOA and the police forces they monitor is warranted. For now, I will conclude with some recommendations I have formed for the CPOA to better mitigate the influence of biases in their investigations and decision-making processes.

Recommendations for the CPOA

There are a few steps that I would recommend the CPOA take to hedge the influence of human bias in their organization and improve their data reporting practices.

- Add multicultural engagement training and sociological theory to board member educational requirements.

  Though hovering in and out of the significance level of $p<0.1$, the indicator of biased-based policing allegations having a negative coefficient in relation to the odds of sustained findings is concerning. Supplementing the many hours of training CPOA board members spend in APD settings with some trainings in community engagement and sociological histories of the American policing systems more broadly could help board members better empathize with communities that have a history of discrimination by police, and thus mitigate the influence pro-police bias.

- Increase community outreach for board members.

  Similar to my first recommendation, I suggest that board members be present for a portion of in-person complaints made to the CPOA (wherever they are most frequently received) such that they might better understand the civilians they are tasked with representing.

- Train complaint recording and processing staff not to mention complainant's mental health conditions in complaint text.

  Some complaint texts from this dataset are poorly authored and sometimes even include personal perspectives of the CPOA employees recording the complaint. This should be guarded against much more carefully, as the complaint text is meant to be representative of the complainant's claims rather than the opinions of CPOA administrators. Employees ought also to avoid direct mention of the complainant's mental health history unless specifically relevant to a claim (e.g. "the officer discriminated against me because he knew I had XYZ mental condition").

- Conduct thorough digital censorship on each case prior to publishing the findings.

  Errors in the redaction of personally identifiable information of complainants in this dataset was shockingly frequent. I would recommend the CPOA develop systematic methods of standardized digital censorship of case information such that the unredacted letter may be sent directly to the complainant, while the censored version is released to the public. The CPOA may have such methods in place for protecting officer identities, as the names of officers appeared much less frequently. Consistency in this matter is imperative.

- Standardize the formatting and reporting of variables, policies reviewed, and nonconcurrence.

First, the CPOA should be more consistent in reporting explanations of their findings, as a non-negligible portion of the finding letters included little to no clarification on the rationale behind their decision to find complaints "unfounded" or "administratively closed." Additionally, the choice in variables reported in a format separate from unstructured text varied over the years. The layout and content of the additional variables should be more consistent. I would also recommend including an indicator of APD leadership nonconcurrence, perhaps even with a short description of their alterations to the discipline recommended by the CPOA. Letters released without clarity on the APD's final disciplinary decisions could be misleading to civilian complainants, and waiting the thirty-day period for nonconcurrence reporting by APD leadership would not constitute a significant delay in the finding letter publication, which averaged 158 days from complaint date.

6.2 Limitations

This project comes with limitations. First, it is important to note that this is an exploratory study designed to investigate the substance of civilian complaints and their results as mediated by the CPOA. Associations between parameters in the dataset are moderated by Tesseract's OCR performance, regular expression captures, and the accuracy of classifier models charged with identifying the prevalence of said variables. Moreover, this project employs strictly open-source tools. The use of a cost-protected LLM or neural network may have produced more efficient and accurate outcomes. The focus of the project on Albuquerque, the CPOA and APD, likewise limits the generalizability of results, and places the research in direct reliance on the city's transparent provision of materials. I had no way of identifying whether there may be materials withheld from online publication, and if CPOA, the city of Albuquerque, or APD leadership chose to systematically exclude any particular case types, the results of this investigation would be impacted. The choice of sampling 10% of the dataset for hand-coding, while defensible given the varying size of finding letters and the time constraints of the project, limited many of my analyses. A larger sample size of hand-coded data would have improved the legitimacy of the findings. Finally, changes in the format of these letters over time also made certain aspects of this project particularly challenging, and likely introduced error into regular expression captures and encoder weightings alike. While I attempted to be as methodical as possible, the supervision necessary to ensure no impact by alteration of letter structure over time would make the use of machine learning models as factor predictors across cases entirely superfluous and hand-coding the entire corpus would prove more efficient.

6.3 Contribution

This study took a solid first step into applying trained machine learning models on textual data in an effort to qualify the output of civilian review boards (CRBs). The existing literature to date showed high potential for the employment of NLP tools in police science, but none thus far had applied it to the context of the CRB. Further, the process showed concerning preliminary evidence of the CPOA's tendency for lenience toward APD employees. Preprocessing of the dataset revealed a variety of challenges facing the use of NLP tools in processing civilian complaint texts including the variation in their format across CRB leadership, the presence of OCR artifacts as a result of inconsistent censorship practices, and the challenge of reliably classifying complaints regarding various items considered "property." Also resulting from this dissertation are recommendations for the CPOA listed above, and directions for future research, included below.

7 | Future Research

While this project serves as a strong initial examination of CPOA activities, there are a few adjustments that might be made in future explorations of the data.

- Refining certain parameters to clarify the boundary between positive and negative classes.

  Removing OBRD use and self-identification from `regulatory_complaint` criteria and focusing the parameter strictly on the content and timing of officers' written reports might capture more specific, interpretable content. Adjusting the `traffic_scenario` variable to include officers running plates for the execution of warrants and adding a separate variable that covers strictly complaints in the context of collisions could also improve the dataset.

- Including a variable (or two) that captures dissonance between a complainant's expectations for officers to press (or not press) charges.

  A term that describes complaints alleging that "the officer pressed charges (s)he shouldn't have" or "the officer didn't press charges (s)he should have" might narrow the portion of the dataset uncaptured by variables of interest.

- Increasing supervision of models.

  Future studies might consider adding an "attention head" unique to the dataset for the SBERT embedding development (Jurafsky and Martin, 2025), or employ a method similar to the "GATE" framework used in studies like Karystianis et al. (2018) to improve the reliability of classification.

- Making cross-agency comparisons.

  Future researchers interested in comparing CRB content across contexts might identify CRBs in America that are positioned similarly to the CPOA in Albuquerque and utilize the methods of this project in extracting pertinent information from all CRBs of interest.

7.1 Ethical Statement

No ethics board approved this study, but unidentified human data was utilized. Uncensored human data was subsequently censored and is not reported here.

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